

# OrangeHRM Capstone Testing Plan

QA automation capstone scope document using Playwright with JavaScript.

| Field               | Details                                                                              |
|---------------------|--------------------------------------------------------------------------------------|
| Project Type        | Enterprise HRM automation testing capstone                                           |
| Selected Website    | OrangeHRM Demo                                                                       |
| Website URL         | https://opensource-demo.orangehrmlive.com/                                           |
| Automation Stack    | Playwright with JavaScript                                                           |
| Minimum Requirement | 8 services, 15 test cases per service, total 120 test cases                          |
| Testing Focus       | Authentication, employee workflows, dashboard validation, API testing, and reporting |

## Project Overview

This capstone project focuses on automating OrangeHRM enterprise workflows using Playwright and JavaScript. The framework follows enterprise automation standards including Page Object Model (POM), reusable utilities, parallel execution, API mocking, reporting, and CI/CD integration.

## Project Objectives

- Prepare and automate 120 test cases across 8 business modules.
- Implement reusable and scalable Playwright framework architecture.
- Perform cross-browser execution using Chromium, Firefox, and WebKit.
- Generate execution evidence through screenshots, videos, traces, and reports.
- Validate UI workflows, API behavior, and data-driven scenarios.

## Technology Stack

- Playwright v1.4x
- JavaScript / Node.js
- Git & GitHub
- Faker.js
- dotenv
- Allure Report
- GitHub Actions

## Modules Covered

1. Authentication & Security
2. Employee Management (PIM)
3. Recruitment
4. Dashboard Analytics
5. Leave Management

- 6. My Info Profile
  - 7. Time Tracking
  - 8. Admin Panel & API Mocking
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## Severity Levels

- Critical: Login failures, application crashes, blocked workflows.
  - High: Broken business flows, API failures, incorrect validations.
  - Medium: UI inconsistencies, data mismatch, alignment issues.
  - Low: Minor styling issues, tooltip and display problems.
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## Automation Approach

The automation framework will use Playwright Test Runner with JavaScript. Tests will be organized module-wise using reusable Page Object classes, fixtures, helpers, and test data files. Reporting will be integrated with HTML and Allure Reports.

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## Framework Structure

```
tests/  
pages/  
fixtures/  
utils/  
api/  
reports/  
playwright.config.js  
package.json
```

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## Execution Strategy

- Execute tests in parallel across multiple browsers.
  - Capture screenshots and traces for failed scenarios.
  - Use stable locators such as getByRole and data-testid.
  - Avoid hard waits and implement dynamic synchronization.
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## Expected Deliverables

- Capstone Testing Plan PDF
  - 120 Automated Test Cases
  - Playwright Automation Framework
  - Allure & HTML Reports
  - Screenshots and Trace Evidence
  - Final Project Documentation
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